

Nanostructured polymers: Macromolecular engineering

Prof. Dr. Sagrario PASCUAL
Le Mans Université (FRANCE)

E-mail: Sagrario.pascual@univ-lemans.fr



Nanostructured polymers: Macromolecular engineering

Outline

- Chapter 1: Introduction to Nanostructured polymers
- Chapter 2: Anionic living polymerization
- Chapter 3: Controlled radical polymerization: From conventional free radical polymerization to reversible-deactivation radical polymerization (RDRP)
- Chapter 4: Controlled radical polymerization: The NMP, ATRP, and RAFT processes
- Chapter 5: Ring-opening metathesis polymerization (ROMP)
- Chapter 6: Macromolecular engineering: Strategies and methods for the synthesis of nanostructured polymers (functional polymers, block and graft copolymers, hyperbranched polymers, dendrimers...)
- Chapter 7: Nanostructured polymers - case studies: Block and graft copolymers.

Chapter 1

Introduction to Nanostructured Polymers

- Nanomaterials
- Polymers as building block for nanomaterials
- Block copolymers in Bulk
- Block copolymers in Solution

Nanomaterials: Definitions

Nanomaterials are **nanoscale materials**, *i.e.*, **materials at a scale below 100 nm**, according to the definition of the International Organization for Standards (ISO) (ISO/TS 80004-2:2008, 2008) **and may exhibit different physical and chemical properties compared with the same substances at normal scale**, such as increased chemical reactivity greater surface area (ISO/TS 80004-1:2010, 2010).

Nanomaterials: Definitions

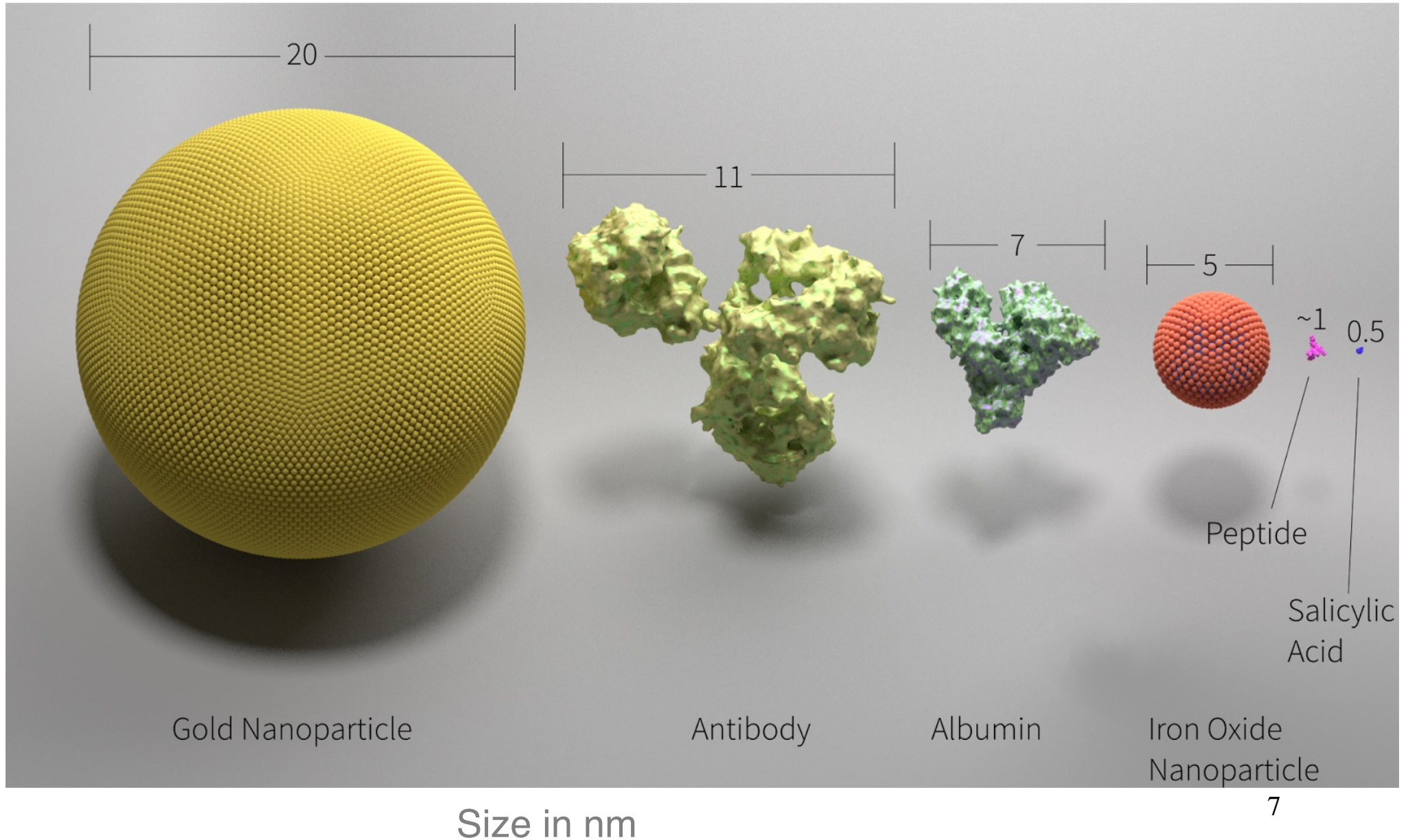
In ISO/TS80004, **nanomaterial** is defined as the "*material with any external dimension in the nanoscale or having internal structure or surface structure in the nanoscale*", with **nanoscale** defined as the "*length range approximately from 1 nm to 100 nm*".

This includes both **nano-objects**, which are discrete pieces of material, and **nanostuctured materials**, which have internal or surface structure on the nanoscale; a nanomaterial may be a member of both these categories.

Nanomaterials: Definitions

In 2011, the European Commission defined a nanomaterial as "**A** *natural, incidental or manufactured material containing particles, in an unbound state or as an aggregate or as an agglomerate and for 50% or more of the particles in the number size distribution, one or more external dimensions is in the size range 1 nm – 100 nm [...].*"

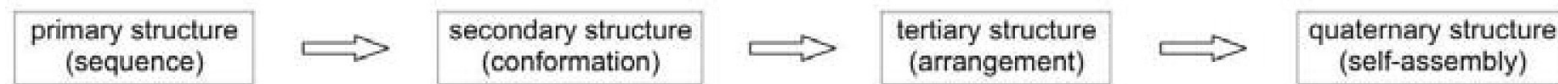
A Matter of Size...



Consecutive organizational levels in natural & synthetic nanoconstruction using macromolecules

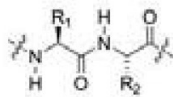
Nature has developed a hierarchical strategy to construct sophisticated and **complex nanostructures**; **Chemistry** explores **synthetic macromolecules** as a means to synthesize specific nano-objects:

Hierarchical structure formation:



- discrete natural biomacromolecules

Natural



polypeptide



α -helix



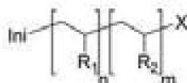
myoglobin



hemoglobin

- polydisperse artificial macromolecules

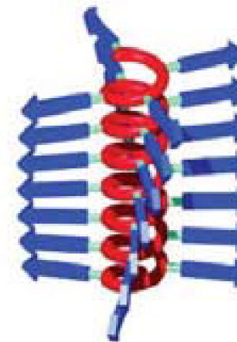
Synthetic



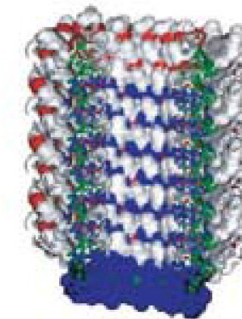
polyolefin
block copolymer



helical isotactic
poly(propylene)



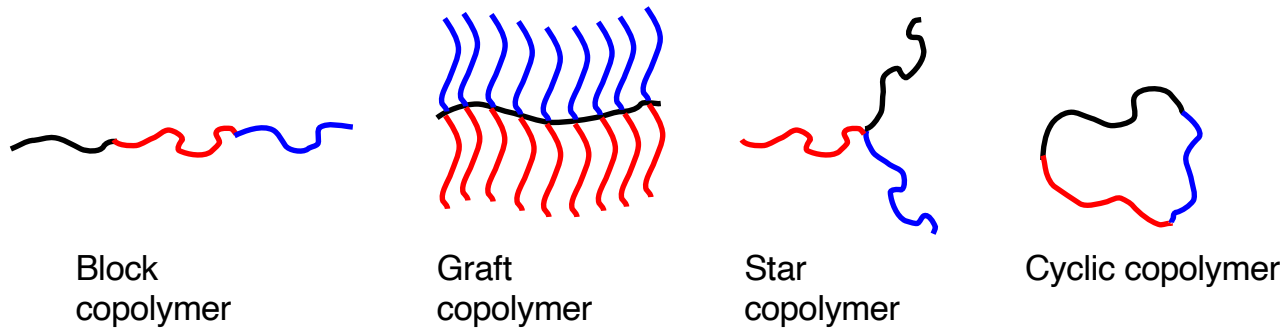
β -helical
poly(isocyanide)



self-assembled
dendritic dipeptide pore

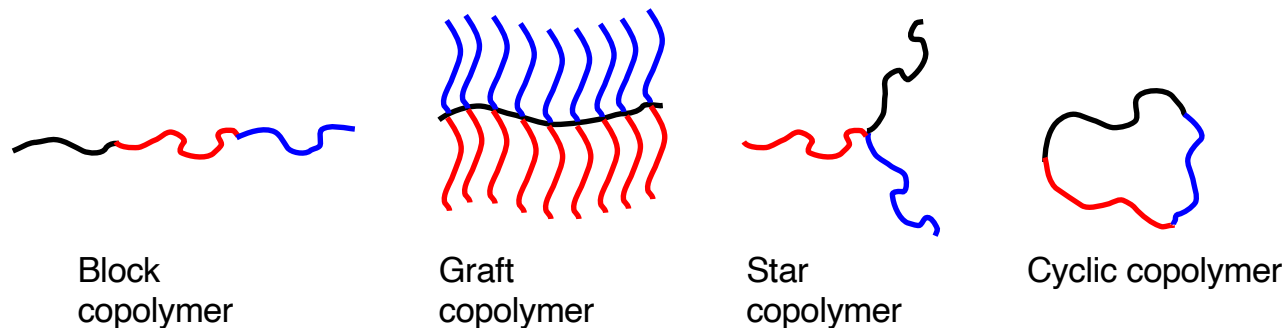
Polymers as building blocks for Nanomaterials

Polymers represent **ideal nanoscale tools**, not only due to their intrinsic **nanoscopic dimensions**, rich chemical variability, ease of synthesis and processing, but also because of their peculiar **phase separation** both in bulk and in solution, particularly in the case of **copolymers** having different structural segments.



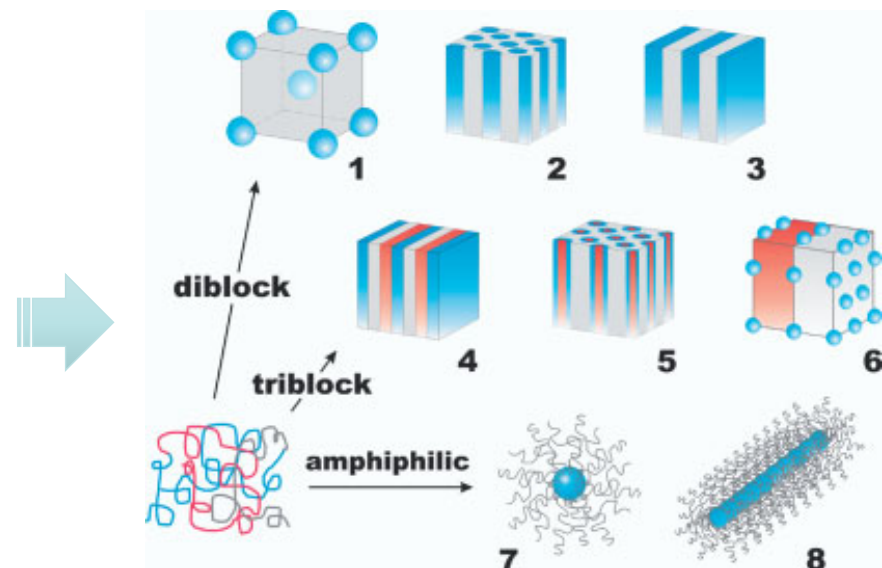
Polymers as building blocks for Nanomaterials

Polymers represent **ideal nanoscale tools**, not only due to their intrinsic **nanoscopic dimensions**, rich chemical variability, ease of synthesis and processing, but also because of their peculiar **phase separation** both in bulk and in solution, particularly in the case of **copolymers** having different structural segments.



By **tuning copolymer properties** such as the chemical structure of the monomers, the block molecular weights, and the copolymer architecture, a suite of micellar structures can be achieved.

In particular, **block copolymers** readily undergo **microphase separation in selective solvents** to form **micelles**, thus affording the opportunity to achieve discrete nanostructures through a spontaneous process.



Polymers as building blocks for Nanomaterials

Copolymers self-assembly is a self-organizing process driven by **noncovalent inter-actions** (*i.e.*, hydrophobic, van der Waals, π – π , electrostatic, hydrogen bonding, etc.) that leads to the formation of stable compartmentalized supramolecular aggregates.

Bond Energy and Relative Strength of Different Intermolecular Forces*

Type of interaction	Bond energy (kJ/mol)	Relative strength
Ionic attraction	850–1700	1000
Hydrogen bonding	50–170	100
Dipole-dipole interaction	2–8	10
van der Waals interaction	~1	1

*Yang & Han, *Macromolecules* **2008**, 41, 2104

Polymers as building blocks for Nanomaterials

The goal of any synthetic method for nanomaterials is to yield a material that exhibits properties that are a result of their characteristic length scale being in the nanometer range (1 – 100 nm).

Accordingly, the **synthetic method** should exhibit control of size in this range so that one property or another can be attained.

Polymers as building blocks for Nanomaterials

The goal of any synthetic method for nanomaterials is to yield a material that exhibits properties that are a result of their characteristic length scale being in the nanometer range (1 – 100 nm).

Accordingly, the **synthetic method** should exhibit control of size in this range so that one property or another can be attained.

The methods are divided into **two main types**:

- **Top down**: uses some force (e. g. mechanical force, laser) to break bulk materials into nanoparticles;
- **Bottom up**: involves the assembly molecules (or atoms) into nanostructured arrays.



*This is the method under
consideration here
with nanostructured polymers.*

Polymers as building blocks for Nanomaterials:

Block Copolymers in Bulk

- **Block copolymers** (BCPs) are macromolecules obtained from two or more blocks of different monomers that are covalently connected to each other. BCPs have generated significant interest due to their **highly tunable nanoscale self-assembly**.

Polymers as building blocks for Nanomaterials:

Block Copolymers in Bulk

- **Block copolymers** (BCPs) are macromolecules obtained from two or more blocks of different monomers that are covalently connected to each other. BCPs have generated significant interest due to their highly tunable nanoscale self-assembly.
- BCPs Interactions between the blocks generally render them **incompatible**.
- This incompatibility will drive a microphase separation process, as bulk phase separation is prohibited by the presence of covalent bonds. Therefore, polymer chains **self-assemble** into **nanoscopic ordered domains** with a **morphology** that depends on the **ratio of block lengths, strength of interaction, and number of blocks**.

Polymers as building blocks for Nanomaterials:

Block Copolymers in Bulk

- **Block copolymers** (BCPs) are macromolecules obtained from two or more blocks of different monomers that are covalently connected to each other. BCPs have generated significant interest due to their highly tunable nanoscale self-assembly.
- BCPs Interactions between the blocks generally render them incompatible.
- This incompatibility will drive a microphase separation process, as bulk phase separation is prohibited by the presence of covalent bonds. Therefore, polymer chains self-assemble into nanoscopic ordered domains with a morphology that depends on the ratio of block lengths, strength of interaction, and number of blocks.
- In a typical BCP, the **dimension of the domains range from 10 nm to 100 nm** and can be **controlled** by changing the overall **molecular weight** of the macromolecule.

Polymers as building blocks for Nanomaterials:

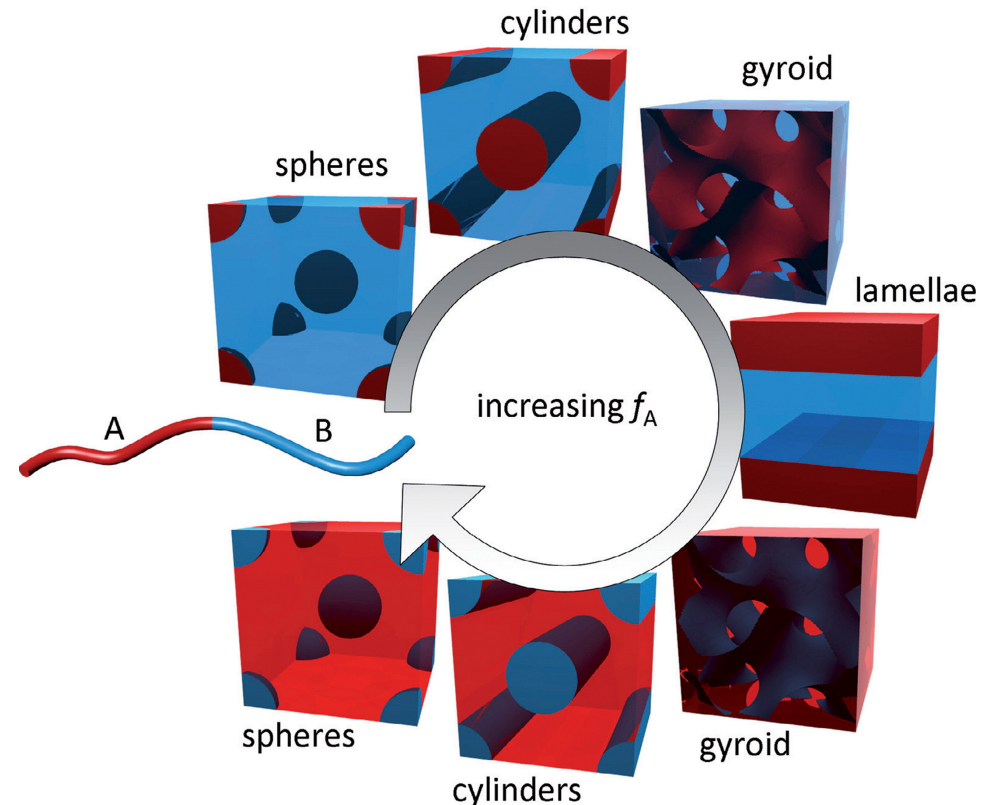
Block Copolymers in Bulk

- **Block copolymers** (BCPs) are macromolecules obtained from two or more blocks of different monomers that are covalently connected to each other. BCPs have generated significant interest due to their highly tunable nanoscale self-assembly.
- BCPs Interactions between the blocks generally render them incompatible.
- This incompatibility will drive a microphase separation process, as bulk phase separation is prohibited by the presence of covalent bonds. Therefore, polymer chains self-assemble into nanoscopic ordered domains with a morphology that depends on the ratio of block lengths, strength of interaction, and number of blocks.
- In a typical BCP, the dimension of the domains range from 10 nm to 100 nm and can be controlled by changing the overall molecular weight of the macromolecule.
- Modern synthetic chemistry provides possibilities to design BCPs with specific length scales and geometries. Such systems can **self-assemble** in a hierarchical way on multiple length-scales ranging from nanometers to macroscopic sizes leading to a vast variety of (idealized) **nanostructures** that can target, in addition to fundamental studies, many applications: nanolithography, controlled drug delivery, organic photovoltaics (OPVs), organic light emitting diodes (OLEDs), etc.

Polymers as building blocks for Nanomaterials: Block Copolymers in Bulk

The **primary ordered phases** observed in **coil-coil diblock copolymers** comprised of two incompatible and flexible blocks, A and B (f_A = volume fraction of block A) in the bulk:

- Body-centered cubic **spheres**
- Hexagonally packed **cylinders**
- Double **Gyroid**
- **Lamellae**.



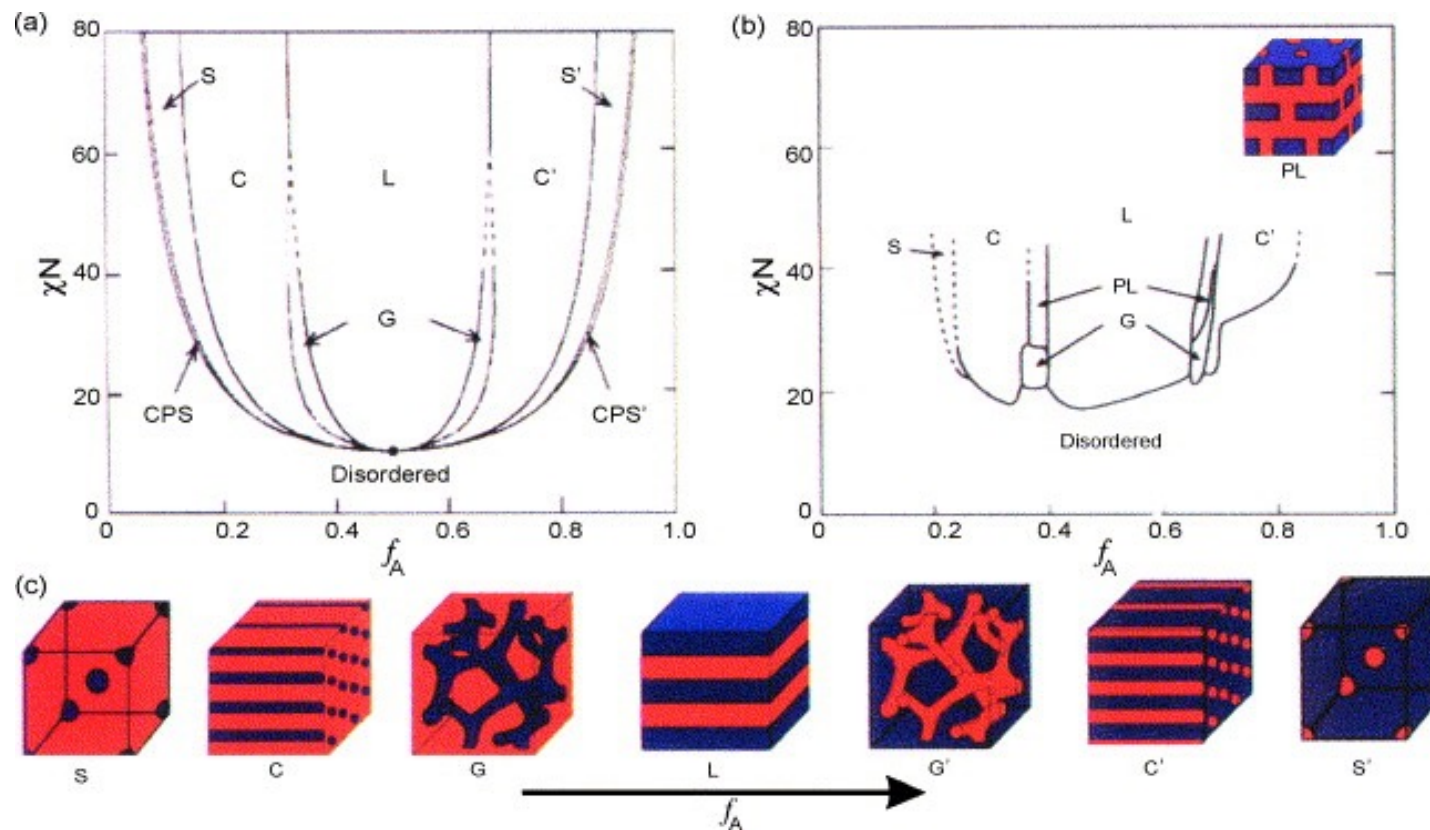
I. Botiz and S. B. Darling, *Mater. Today*, **2010**, 13, 42–51.

This phase diagram will change when **coil blocks are replaced by blocks that are semi-flexible or rigid rods**, as is the case with most conjugated polymers. Moreover, introduction of crystallizable blocks leads to a competition between crystallization and microphase separation, which can substantially affect the morphology.

Polymers as building blocks for Nanomaterials: Block Copolymers in Bulk

Typical phase diagram of a **coil-coil diblock** copolymer **PolyA-*b*-PolyB**

Diblock copolymers are predicted to self-assemble according to a phase diagram predicted by self-consistent mean field theory (a) and proven experimentally (b):



f_A : Volume fraction of block A. χ : Flory-Huggins interaction parameter. N: degree of polymerization.
S: body-centered spheres, CPS: closed-packed spheres, C: hexagonally packed cylinders, L: lamellae, G: double gyroid phase.

Polymers as building blocks for Nanomaterials:

Block Copolymers in Solution

Micellization of Block Copolymers

- Block copolymers self-assemble into aggregates in specific solvents at a certain temperature **above** their critical micelle concentration (**CMC**).
- The simplest micelles are spherical. Other morphologies include rods, bicontinuous structures, lamellae, tubules, and several others.

Polymers as building blocks for Nanomaterials:

Block Copolymers in Solution

Micellization of Block Copolymers

- Block copolymers self-assemble into aggregates in specific solvents at a certain temperature **above** their critical micelle concentration (**CMC**).
- The simplest micelles are spherical. Other morphologies include rods, bicontinuous structures, lamellae, tubules, and several others.
- **Critical micelle concentration (CMC):** Below the CMC, block copolymers in solution behave as **unassociated unimers** (*i.e.*, no self-assembly). **Above the CMC**, block copolymers **self-assemble** into **micelle structures**.
- The CMC is dependent on several factors, including copolymer type, molecular weight of each polymer block, concentration, temperature, solvent, and pH value.

Polymers as building blocks for Nanomaterials:

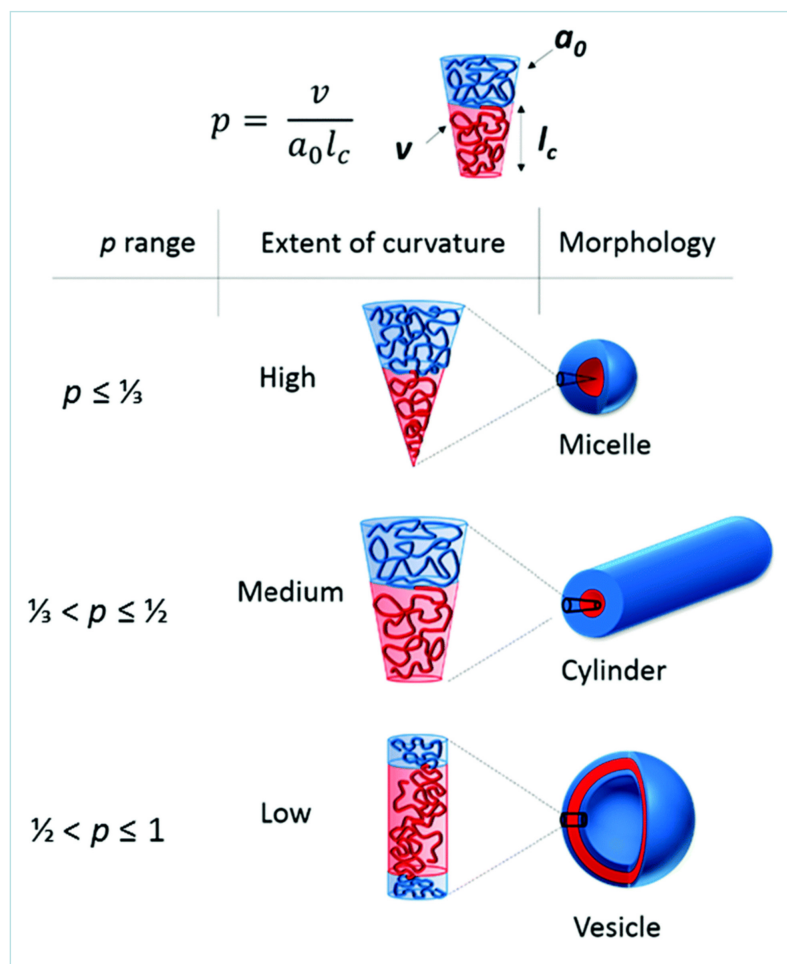
Block Copolymers in Solution

Micellization of Block Copolymers

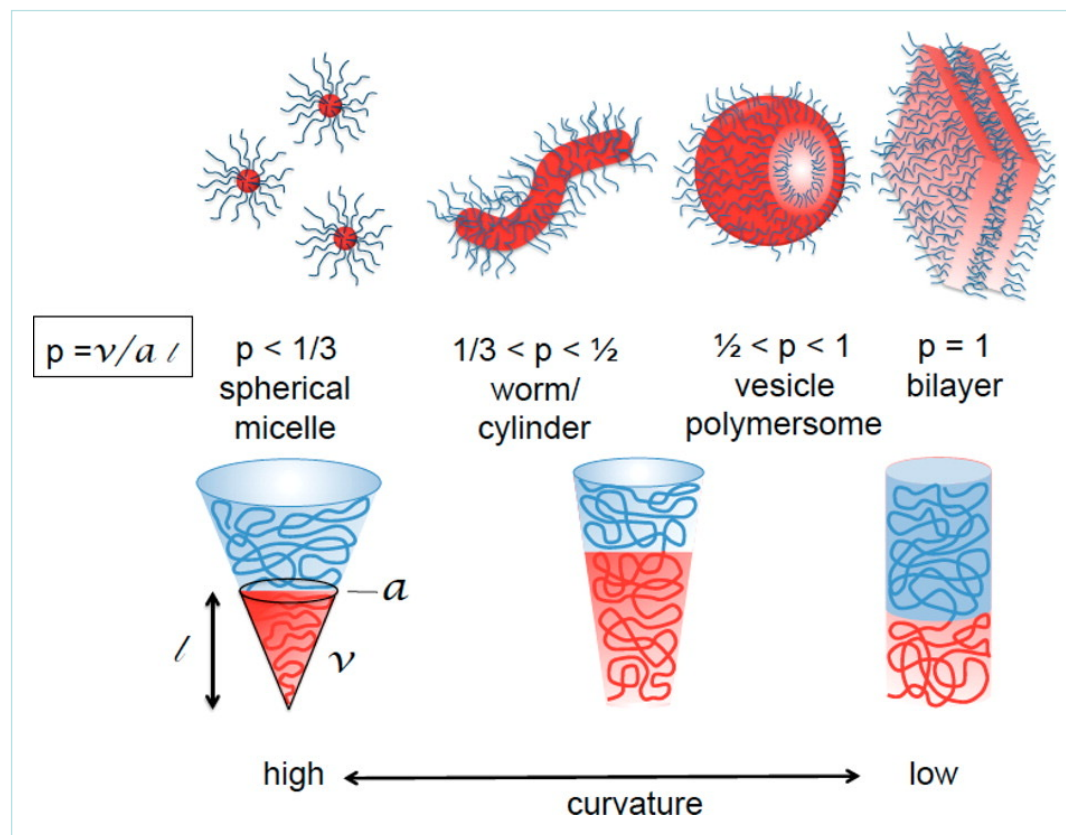
- Block copolymers self-assemble into aggregates in specific solvents at a certain temperature **above** their critical micelle concentration (**CMC**).
- The simplest micelles are spherical. Other morphologies include rods, bicontinuous structures, lamellae, tubules, and several others.
- Critical micelle concentration (CMC): Below the CMC, block copolymers in solution behave as unassociated unimers (*i.e.*, no self-assembly). Above the CMC, block copolymers self-assemble into micelle structures.
- The CMC is dependent on several factors, including copolymer type, molecular weight of each polymer block, concentration, temperature, solvent, and pH value.
- **Aggregation or association number, Z** : The number of block copolymers within the micelle. For diblock copolymers, this is generally defined by: $Z \approx N_B^\alpha N_A^{-\beta}$, where N_A and N_B are the number of repeat units of insoluble and soluble polymer blocks, and α and β are the scaling exponents of the two blocks, respectively.

Polymers as building blocks for Nanomaterials: Block Copolymers in Solution

Block copolymer morphologies in solution: The morphology in relatively diluted solution is predicted using the **packing parameter** “ p ”:



$p = v/A_l$ with v = volume of block A, l = length of block A
 a = effective interfacial area of the block junction



Polymers as building blocks for Nanomaterials:

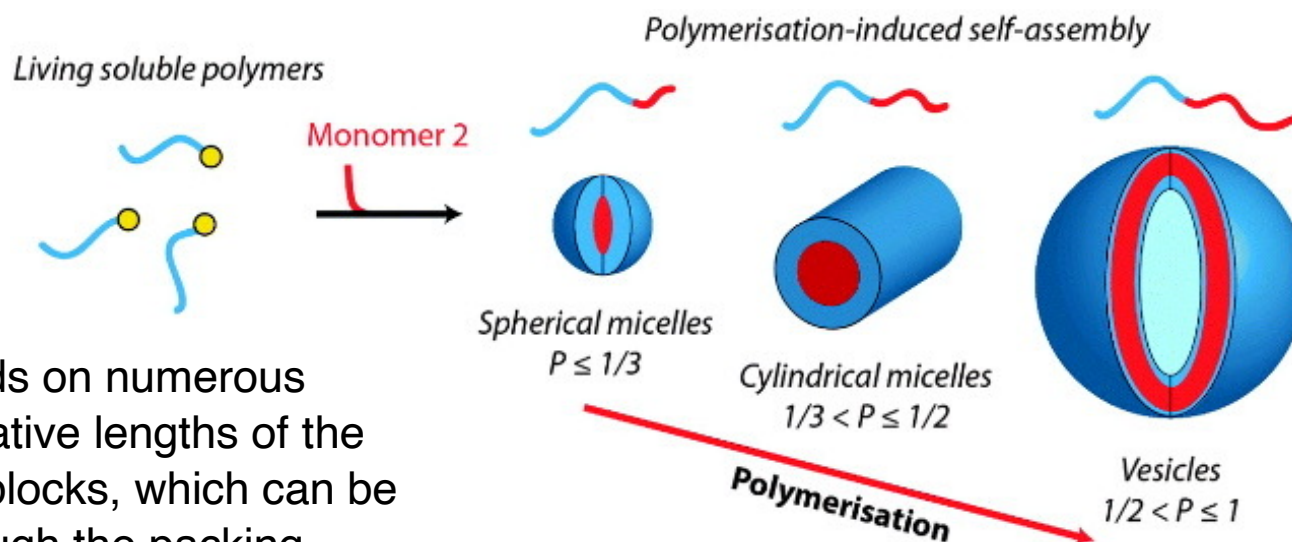
Block copolymers in Solution - PISA

Polymerisation-Induced Self-Assembly (PISA) of amphiphilic block copolymers: An efficient route to nanostructured materials

Controlled synthesis of block copolymers has been applied in heterogeneous systems, motivated by the numerous process advantages and the position of emulsion polymerisation at the forefront of industrial latex synthesis.

In addition to the inherent environmental advantages of heterogeneous routes, the incidence of **block copolymer self-assembly** within dispersed particles **during polymerisation** (**polymerisation-induced self-assembly, PISA**) leads to novel nanostructured materials:

Schematic of the **PISA** of **amphiphilic block copolymers** upon polymerisation in a selective solvent.

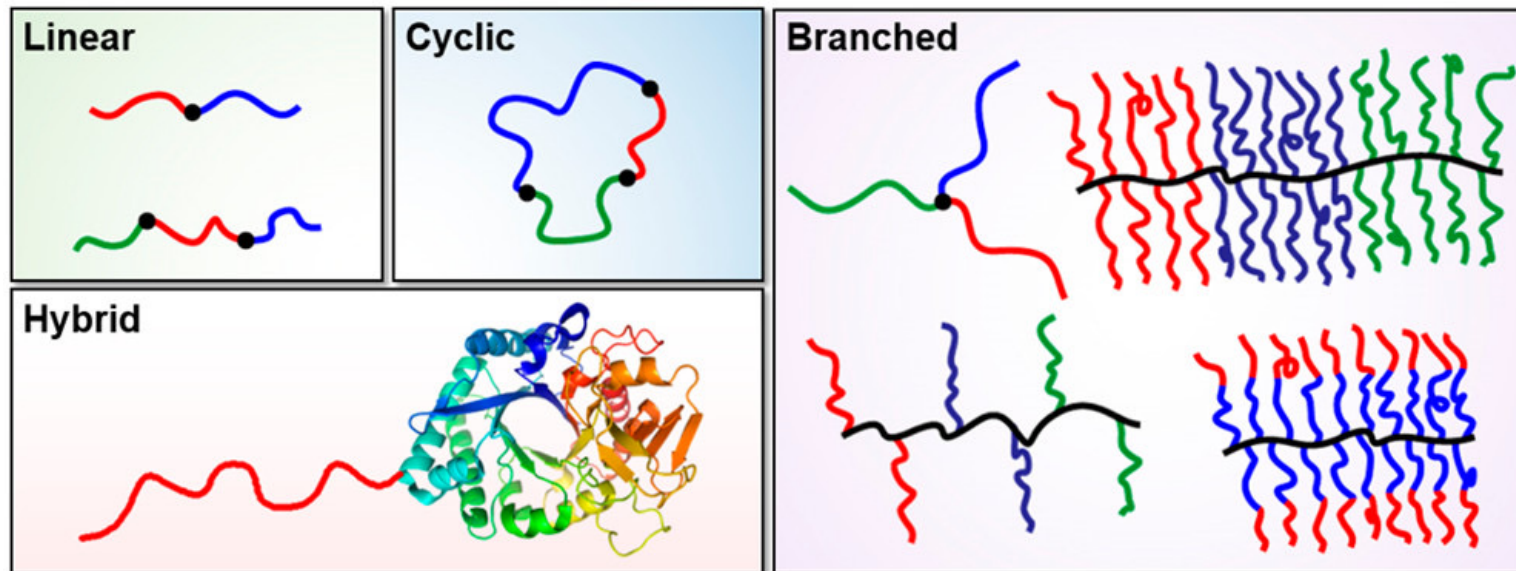


The **final morphology** depends on numerous parameters, primarily the relative lengths of the **solvophilic** and **solvophobic** blocks, which can be quantitatively estimated through the packing parameter (P).

Polymers as building blocks for Nanomaterials

From **Macromolecules** to **Nanomaterials**:

Macromolecular engineering based on **modern synthetic chemistry** methods offers possibilities to design polymers with specific composition, dimensions, and topology.



*The methods of **macromolecular engineering** will be the subject of the next lectures...*

Nanostructured polymers: Macromolecular engineering

Outline

- Introduction to Nanostructured polymers
- Anionic living polymerization
- Controlled radical polymerization: From conventional free radical polymerization to reversible-deactivation radical polymerization (RDRP)
- Controlled radical polymerization: The NMP, ATRP, and RAFT processes
- Ring-opening metathesis polymerization (ROMP)
- Macromolecular engineering: Strategies and methods for the synthesis of nanostructured polymers (functional polymers, block and graft copolymers, hyperbranched polymers, dendrimers...)
- Nanostructured polymers - case studies: Block and graft copolymers.

Bibliography

Edited by Krzysztof Matyjaszewski,
Yves Gnanou, Ludwik Leibler

WILEY-VCH

Macromolecular Engineering

Precise Synthesis, Materials Properties, Applications

Volume 1: Synthetic Techniques

